DEPARTMENT OF INFORMATION SYSTEMS

INF2011S – Systems Design & Development

Future Path

Student Advising System

Case Study and Test Data Specification

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PART A

Future Path Student Advising System Case Study

1. Business Background

Many Grade 11 and Grade 12 learners in South Africa face uncertainty when deciding what to study after school. Learners are often interested in particular degrees or career paths but may not know whether their current subjects, marks and interests align with the requirements of those qualifications. In many cases, learners only discover that they do not meet the minimum requirements when they are already close to applying, or after they have already submitted an application.

This problem is especially common among learners who do not have regular access to career advisors, university open days or detailed admission information. Some learners rely on parents, teachers, friends, social media or outdated information when making important study decisions. This can result in unrealistic study choices, missed funding opportunities, late applications and poor alignment between the learner's strengths and their intended field of study.

To support learners earlier in their decision-making process, a new standalone Information System has been proposed. The system will be called Future Path. Future Path is a student pathway advising system designed to help Grade 11 and Grade 12 learners explore possible study options, check whether they appear to meet the minimum criteria for selected degrees, identify gaps in their current academic profile, view relevant funding information, and export an advisory summary that can be shared with a parent, teacher or career advisor.

The system is not intended to make official admission decisions. Instead, it provides preliminary guidance based on locally stored degree information, subject requirements, interest categories, funding options and rule-based recommendation logic.

2. Strategic Purpose of Future Path

The purpose of the system is to provide learners with a structured and accessible way to understand their possible study pathways before they formally apply to university. The system aims to:

- help learners understand whether their school subjects and marks align with selected qualifications;
- suggest alternative degrees where the learner's current academic profile may be better suited;
- match learner interests to faculties, degrees and career areas;
- identify subject or mark gaps that the learner may need to improve;
- provide faculty- or degree-related funding information;
- generate an exportable advisory report that can be shown to parents, teachers or career advisors;
- support more informed, realistic and timely study decisions.

The system will be developed as a standalone **Windows Forms-based desktop application** with a **local relational database**. It must follow a three-tier architecture consisting of a presentation layer, business logic layer and data layer. The system must not rely on a web server, cloud service, internet connection or external APIs.

3. Operating Details

Future Path will be used by learners, school career advisors and system administrators.

Learners will enter their current school grade, subjects, marks, areas of interest and preferred faculty or qualification. Future Path will then use locally stored rules and data to generate an advisory outcome. This outcome may indicate that the learner is a strong match, a possible match, or not currently eligible for the selected qualification.

For example, a learner may be interested in a Bachelor of Commerce degree but may not know whether Mathematics is required, whether Mathematical Literacy is accepted, or what level of performance is expected. Another learner may be interested in a Science-related qualification but may not know whether Physical Sciences is required. Future Path should guide learners by comparing their information against the qualification requirements stored in the system.

Future Path should also allow learners to complete a short interest questionnaire. The learner's responses will be used to calculate an interest alignment score. This score will help the system suggest study areas that may be suitable for the learner's interests.

Future Path should also ask whether the learner would like more information about possible funding options. If the learner selects this option, the system should display funding information linked to the selected faculty, qualification or study field. This may include scholarships, bursaries, financial aid options, merit awards, or other sample funding opportunities stored in the database.

Once the learner has completed the advising process, the system should allow the learner to export a summary report. This report should include the learner's selected degree, subject and mark summary, eligibility outcome, interest match result, identified gaps, possible alternative qualifications, and relevant funding options.

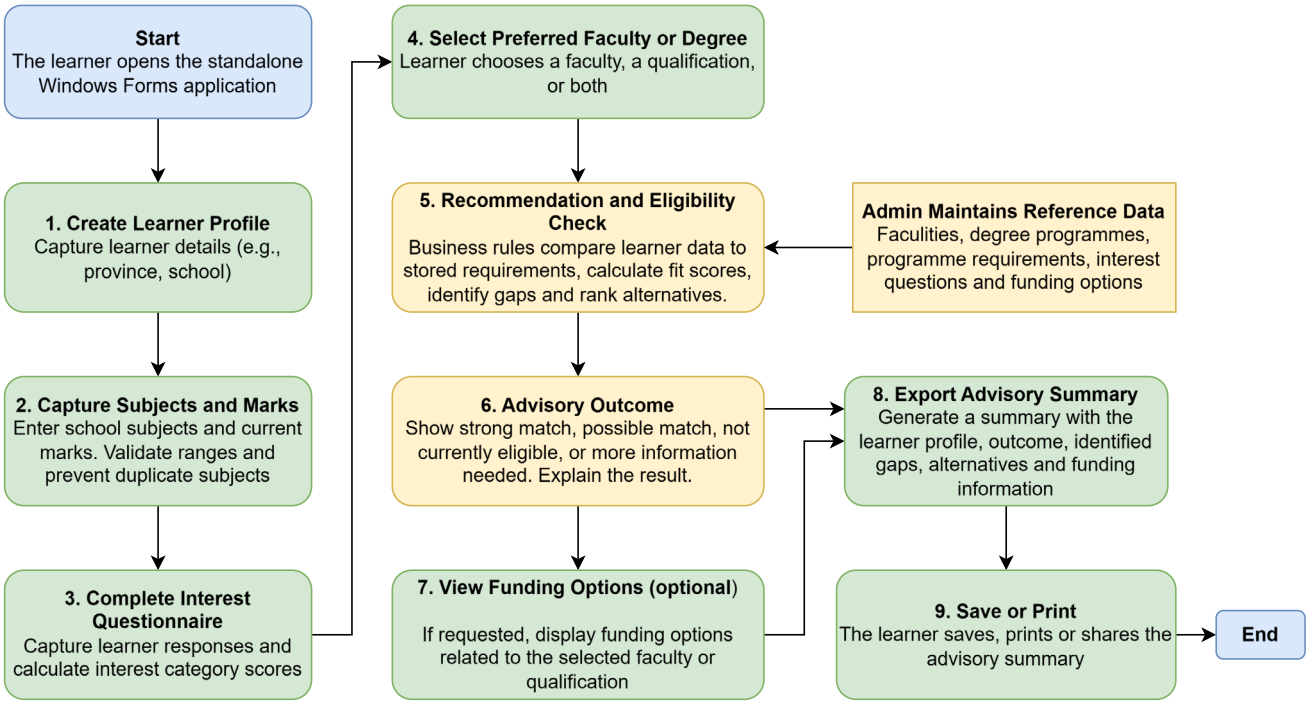


Figure 1: High-Level System Process Flow for Future Path Student Advising System

4. Important System Constraint

Future Path must include recommendation and advising functionality. However, the system must not use ChatGPT, Gemini, Copilot, online APIs, cloud-based AI tools or any external services.

The recommendation and advising functionality must be implemented using local system logic, such as:

- rule-based eligibility checking;
- keyword or category matching;
- interest scoring;
- weighted recommendation scores;
- subject and mark comparison;
- ranking of possible degrees;
- gap analysis;
- database-driven advisory messages.

The system should therefore simulate an intelligent advising experience while remaining fully standalone and database-driven.

5. Existing Problem Context

Currently, many learners make study choices using fragmented information. A learner may check a university website, ask a teacher, speak to a parent, attend an open day, or rely on advice from peers. However, this process is often inconsistent. Learners may not understand how minimum requirements work, may confuse admission requirements with selection likelihood, or may overlook funding opportunities.

Schools may also have limited career guidance capacity. One teacher may be responsible for advising many learners, and it may not be practical to manually assess each learner's subjects, marks, interests and possible degree options. Learners may also be embarrassed to ask questions or may not know what questions to ask.

Future Path is intended to support this process by providing a structured, repeatable and transparent advising tool.

6. Typical Business Processes

6.1 Learner Profile Creation

A learner starts by creating a basic profile. The system should allow the learner to enter their name, current grade, school, province and contact details. The system may also allow the learner to continue without entering sensitive personal information, depending on the design decisions made by the project team.

The learner must indicate whether they are currently in Grade 11 or Grade 12. This is important because Grade 11 learners may be using projected or current marks, while Grade 12 learners may be using final or most recent results.

6.2 Capturing Subjects and Marks

The learner must enter their school subjects and current marks. The system should allow the learner to add multiple subjects, such as Mathematics, Mathematical Literacy, Physical Sciences, Life Sciences, Accounting, Business Studies, Geography, History, English and other relevant subjects.

The system must validate marks to ensure that they are within a valid range. For example, a mark cannot be below 0 or above 100. The system should also prevent a learner from entering the same subject more than once for the same advising session.

6.3 Selecting a Preferred Faculty or Qualification

The learner can either select a preferred faculty first or search for a specific qualification. Faculties may include Commerce, Humanities, Science, Engineering, Health Sciences, Law, Education or other locally stored faculty options.

The learner may select one degree programme for assessment during an advisory session. The system may recommend multiple alternative programmes. The system should retrieve the requirements for the selected degree from the database and compare these requirements with the learner's captured subjects and marks.

6.4 Completing an Interest Questionnaire

The learner should answer a short interest questionnaire. This questionnaire will help the system determine whether the learner's interests align with the selected qualification.

Example interest areas may include:

- working with numbers and data;
- solving technical or scientific problems;
- helping people and communities;
- writing, debating and analysing ideas;
- working with business and entrepreneurship;

- designing systems or technology solutions;
- working in healthcare or life sciences;
- understanding society, politics or law.

Each response should contribute to one or more interest categories. The system should use these responses to calculate an interest alignment score for relevant degrees or faculties.

6.5 Checking Degree Eligibility

The system should compare the learner's subjects and marks against the stored requirements for the selected qualification.

The system should produce one of the following advisory outcomes:

Outcome	Meaning
Strong Match	The learner appears to meet the minimum requirements and has strong interest alignment.
Possible Match	The learner meets some requirements, but has one or more gaps or weaker alignment.
Not Currently Eligible	The learner does not currently meet one or more important requirements.
More Information Needed	The system cannot complete the check because required information is missing.

The system must clearly explain the reason for the outcome. For example, it should not only say "Not Currently Eligible". It should explain that the learner may be missing a required subject or that a mark is below the required threshold.

6.6 Recommending Alternative Degrees

If the learner does not appear to meet the requirements for the selected degree, the system should recommend possible alternatives. These recommendations should be based on the learner's current subjects, marks and interest profile.

For example, if a learner is interested in a qualification that requires Mathematics but only has Mathematical Literacy, the system may suggest alternative qualifications where Mathematical Literacy is accepted, depending on the locally stored sample data.

6.7 Viewing Funding Options

After the system displays the learner's advisory outcome, the learner should be asked whether they would like to view possible funding options related to the selected faculty or qualification.

If the learner selects yes, the system should display relevant funding options from the database.

Funding field	Example
Funding name	Faculty Merit Award
Funding type	Scholarship, bursary, financial aid, external funding
Linked faculty	Commerce
Linked degree	Optional
Eligibility description	Based on academic merit/financial need
Required documents	ID, results, proof of income

Closing date	Stored in database
Contact/URL field	Stored as text, not necessarily live internet

The system does not need to submit funding applications. It only provides locally stored funding guidance.

6.8 Exporting the Advisory Summary

The learner should be able to export an advisory summary. This report should be suitable for sharing with a parent, teacher, school career advisor or for the learner's own planning.

The export summary should include:

- learner profile summary;
- selected faculty or degree;
- captured subjects and marks;
- interest category results;
- eligibility outcome;
- explanation of gaps;
- suggested alternative degrees;
- relevant funding options;
- date generated;
- advisory disclaimer.

The export may be implemented as a printable report, PDF-style report, saved file or on-screen report, depending on the implementation approach selected by the student team.

7. System Users

User	Description
Learner	The learner uses the system to enter their subjects, marks and interests. The learner can view recommendations, funding information and export their advisory summary.
Career Advisor	A career advisor may assist learners in using the system, access learner and advisory-session information, view advisory outcomes and exported summaries, and generate advising and exception reports to support discussions with learners and parents.
System Administrator	The system administrator maintains the reference data in the system. This includes faculties, degrees, admission requirements, interest categories, questionnaire questions and funding options. The administrator should be able to add, update or deactivate outdated degree and funding information.

8. Data Requirements

The system must store enough data to support meaningful advising. The database should include data about learners, subjects, marks, faculties, qualifications, admission requirements, interests, recommendations, funding options and exported reports.

Entity	Description
Learner	Stores learner profile details.
GradeLevel	Stores Grade 11 and Grade 12 categories.
Subject	Stores school subjects.
LearnerSubject	Stores the learner's subject choices and marks.
Faculty	Stores faculties.
DegreeProgramme	Stores qualifications offered by each faculty.
ProgrammeRequirement	Stores subject and mark requirements for each degree.
InterestCategory	Stores broad interest areas.
InterestQuestion	Stores questionnaire questions.
LearnerInterestResponse	Stores learner questionnaire responses.
DegreeInterestMapping	Links degrees to relevant interest categories.
AdvisorySession	Stores each advising session completed by a learner.
RecommendationResult	Stores degree recommendations and scores.
FundingOption	Stores funding opportunities.
FundingFaculty	Links funding options to faculties.
FundingProgramme	Links funding options to specific degrees where relevant.
ExportReport	Stores details of exported learner summaries.
UserAccount	Stores login, role and access details for system users requiring authentication, including Administrators and Career Advisors

The database must be normalised to third normal form. The system should not store repeated groups of subjects, marks, funding options or requirements in a single table field.

Additional data requirements:

- At least five interest categories must be stored.
- Each degree programme must be mapped to at least one relevant interest category.
- At least one active Administrator account and one active Career Advisor account must exist for testing authentication and role-based access.

9. Recommendation Logic

The system should contain a recommendation engine. The recommendation engine does not need to use real artificial intelligence or machine learning. It should be implemented using business rules and scoring logic, comparable to rule-based Expert Systems.

A possible scoring approach is shown below:

Score Component	Description
Subject Match Score	Checks whether the learner has the required subjects.

Mark Match Score	Checks whether the learner's marks meet the stored minimum values.
Interest Match Score	Compares learner interests with degree interest categories.
Faculty Preference Score	Gives additional weight to degrees in the learner's preferred faculty.
Overall Recommendation Score	Combines the above scores into an overall advisory result.

The system should show the recommendation in a way that is understandable to learners. For example:

> ***You are a possible match for this qualification. You meet the general subject requirements, but your Mathematics mark is below the suggested minimum. You may need to improve this mark or consider one of the alternative programmes listed below.***

10. Reports

The system must generate reports that support learner advising, career guidance and system administration. Reports may include summary reports, detailed reports, executive reports, operational reports and exception reports.

At minimum, each project team must design and implement three meaningful reports. These reports should be relevant to the purpose of the system and should use data stored in the local database.

10.1 Report 1: Learner Advisory Summary Report

This report is generated for an individual learner. It summarises the learner's profile, subjects, marks, interest results, selected qualification, eligibility outcome, recommended alternatives and funding options. This report should be exportable so that the learner can share it with a parent, teacher or career advisor.

10.2 Report 2: Degree Interest and Eligibility Report

This report is generated by an administrator or career advisor. It provides a summary of advising activity across learners. The report may include:

- most selected faculties;
- most selected degrees;
- number of learners classified as strong matches;
- number of learners classified as possible matches;
- number of learners not currently eligible;
- common subject gaps;
- common mark gaps;
- most requested funding categories.

This report can help schools or advisors identify where learners need more support.

10.3 Report 3: Learners Not Currently Eligible Exception Report

This report is generated by a career advisor or system administrator. It lists learners whose advisory outcome is Not Currently Eligible or More Information Needed. The report helps identify learners who may need further guidance, subject advice or alternative programme recommendations.

11. Input and Output Controls

The system should include validation and controls to reduce errors.

Examples of input controls include:

- marks must be between 0 and 100;
- required fields must not be left blank;
- a learner cannot enter the same subject twice in one advisory session;
- a degree must belong to a valid faculty;
- a funding option must have a valid funding type;
- closing dates must be valid dates;
- administrator functions must require login.

Examples of output controls include:

- advisory outcomes must include a disclaimer;
- exported reports must show the date generated;
- recommendations must show the reason for the outcome;
- funding information must show whether it is sample information or requires verification;
- inactive degrees or outdated funding options should not appear in learner recommendations.

12. Security and Privacy Considerations

The system should only collect information needed for advising. Since the system is designed for school learners, the project team should consider privacy carefully.

The system should not require unnecessary sensitive personal information. Learner names may be optional if the advisory session is only for guidance. Administrators and Career Advisors must authenticate before accessing functionality associated with their respective roles. Appropriate role-based access restrictions must be implemented.

Administrators should be the only users allowed to maintain degree requirements, funding information and interest questionnaire data.

13. Future Requirements

The System may later be expanded to include additional features, such as:

- support for multiple universities;
- more detailed career pathway mapping;
- parent or teacher login;
- official application checklist generation;
- reminders for application and funding closing dates;
- integration with online university information sources;
- multilingual support;
- improved natural language questioning.

These future requirements are outside the scope of the current student project.For the current project, the system must remain standalone, Windows Forms-based and must use only a local database.

14. Project Task

The analysis phase of the project has been completed. You are now at the design and development phase.

Your task is to design and develop the Future Path Student Advising System according to the requirements described in this case study.

Your solution must:

- be a standalone Windows Forms-based application;
- use a local relational database;
- follow a three-tier architecture;
- include a rule-based recommendation feature using local rules and scoring logic;
- allow learners to capture subjects, marks and interests;
- check selected degree requirements;
- recommend suitable alternatives;
- provide relevant funding information;
- generate an exportable learner advisory summary;
- Include at least three appropriate reports, such as a summary report, detailed report, executive report, operational report or exception report
- Include controls and testing.

The system must not connect to external AI tools, online APIs, cloud platforms or web servers.

PART B

Future Path Student Advising System Test Data Specification

1. General

As each group will design and construct their own domain classes and database, this specification does not define the exact structure of all records in the test database. However, each group must ensure that their database contains enough test data to allow markers to determine whether the system performs as specified.

Future Path should include enough learner, degree, requirement, interest and funding data to test the main system functionality.

2. Learners

The system must contain at least 10 learner records. The following learner records must be included:

Learner	Grade	Subjects and Marks	Preferred Pathway	Expected Outcome
Gadija Jacobs	12	English 72%; Mathematics 70%; Accounting 74%; Business Studies 79%; Economics 70%; Life Orientation 80%	BCom Information Systems	Strong match
Aadiel-Raboo Magwa	12	English 65%; Mathematical Literacy 78%; Business Studies 71%; History 69%; Tourism 75%; Life Orientation 82%	BSc Computer Science	Not currently eligible
Labeeqah Rademeyer	11	English 80%; Mathematics 58%; Physical Sciences 62%; Life Sciences 70%; Geography 66%; Life Orientation 85%	BSc Biological Sciences	Possible match, with recommendation to improve Mathematics
Siyabonga Dickson	12	English 70%; Mathematics 75%; Physical Sciences 78%; Engineering Graphics and Design 73%; Life Orientation 80%	BSc Engineering (Mechanical Engineering)	Strong match
Derek Vin Diesel	12	English 82%; Mathematical Literacy 76%; History 79%; Life Sciences 68%; Visual Arts 85%; Life Orientation 90%	Bachelor of Health Sciences	Not currently eligible, Mathematics and Physical Sciences required.

The remaining learner records may be created by the project team.

3. Subjects

The database must contain at least 16 school subjects, including:

- English;
- Afrikaans;

- Mathematics;
- Mathematical Literacy;
- Physical Sciences;
- Life Sciences;
- Accounting;
- Business Studies;
- Economics;
- Geography;
- History;
- Computer Applications Technology;
- Life Orientation;
- Tourism;
- Engineering Graphics and Design;
- Visual Arts.

4. Faculties

The database must contain at least 5 faculties, for example:

- Commerce;
- Humanities;
- Science;
- Engineering;
- Health Sciences.

5. Degree Programmes

The database must contain at least 10 degree programmes.

Examples may include:

- BCom Information Systems;
- BCom Accounting;
- BCom General;
- BSc Computer Science;
- BSc Biological Sciences;
- BSc Engineering (Mechanical Engineering);
- BA Humanities;
- BSocSci;
- Law-related qualification;
- Bachelor of Health Sciences.

The project team may use fictionalised or sample programme names and requirements. The system must not claim that sample requirements are official current university requirements.

6. Programme Requirements

Each degree programme must have at least one stored requirement.

Requirements may include:

- required subject;
- minimum mark for required subject;
- accepted alternative subject;
- minimum overall average;
- faculty-specific rule;
- advisory note.
- At least three programmes must require Mathematics.
- At least two programmes must require Physical Sciences.
- At least two programmes must accept Mathematical Literacy.
- At least one programme must have requirements that Gadija Jacobs meets.
- At least one programme must have requirements that Aadiel-Raboo Magwa does not meet.
- The BSc Biological Sciences requirements must result in Labeeqah Rademeyer receiving a Possible Match**,** with her Mathematics mark identified as a gap.
- The BSc Engineering (Mechanical Engineering) requirements must result in Siyabonga Dickson receiving a Strong Match**.**
  • The Bachelor of Health Sciences requirements must require Mathematics and Physical Sciences, resulting in Derek Vin Diesel being Not Currently Eligible.

7. Interest Questionnaire

The system must contain at least 8 interest questions. Example questions include:

1. I enjoy solving problems using numbers or data.
2. I enjoy helping people and communities.
3. I enjoy working with technology.
4. I enjoy science experiments and research.
5. I enjoy reading, writing and analysing ideas.
6. I enjoy business, entrepreneurship and finance.
7. I enjoy designing or building practical solutions.
8. I enjoy understanding society, law or politics.

The system must store learner responses and use them to calculate an interest alignment result.

8. Funding Options

The system must contain at least 8 funding option records.

Each funding option should include:

- funding name;
- funding type;
- linked faculty or programme;

- eligibility description;
- required documents;
- closing date;
- contact or application information;
- active/inactive status.
- At least one funding option must be linked to Commerce.
- At least one funding option must be linked to Science.
- At least one funding option must be linked to Engineering.
- At least one funding option must be linked to Health Sciences.
- At least one funding option must be linked to financial need.
- At least one funding option must be linked to academic merit.

9. Advisory Sessions

The system must contain at least 10 advisory session records.

The data should demonstrate the following scenarios:

- a learner who strongly matches their selected degree;
- a learner who does not meet a required subject rule;
- a learner whose mark is below the minimum requirement;
- a learner whose interest profile aligns with a different faculty;
- a learner who requests funding information;
- a learner who exports an advisory report.
- A learner for whom required information is missing, resulting in a More Information Needed advisory outcome.

10. Test Cases

Each group must prepare at least 5 test cases to demonstrate that the system meets the user requirements.

The test cases should include:

1. Create a learner profile and capture subjects and marks.

2. Check eligibility for a selected degree.

3. Generate alternative recommendations for a learner who does not meet requirements.

4. View funding options linked to a selected faculty or degree.

5. Export a learner advisory summary report.

The test cases do not need to be submitted with the system specification, but they must be available during the demonstration and used to show that the system meets the required functionality.